Implementation of Fall Detection for The Elderly Home Support using 24-GHz Radar Sensor

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Abstract— **In this paper, we propose a development of a smart home system for assisting elderly people by implementing a fall detection using 24-GHz radar sensor. The system is mounted on the ceiling of the room and the system records the doppler frequency shift signals of a body moving with a duration of 2.5 seconds. The recorded signals are transformed into spectrogram images with a size of 119 x 65 pixels. Then the spectrogram images are recognized by the Convolutional Neural Network (CNN) algorithm. There are 3 classes to be recognized, namely: falling, vacant, and walking. The dataset was acquired, built, and trained by using Edge Impulse platform. The hardware used in the deployment phase is Arduino Nano 33 BLE Sense. The experimental results show an accuracy rate of the training phase and the testing phase of 98.1% and 96.30%, respectively.**

Keywords—Elderly home support, fall detection, 24-GHz radar sensor, edge impulse

I. INTRODUCTION

Elderly, according to Law Number 13 of 1998 is someone who has reached the age up to 60 (sixty) years old and above. In almost five decades, the percentage of Indonesian's elderly has increased by about twice (1971-2020), that is, to 9.92% (26 million), where the elderly women are about one percent more than the elderly men (10.43% versus 9.42%). The data from the National Socio-Economic Survey of Indonesia (Survei Sosial-Ekonomi Nasional Indonesia or Susenas) in 2020 shows that 9.80% of the elderly live alone, and the percentage of older women living alone is almost three times that of older men (14.13% versus 5.06%) [1].

Falls are an unpredictable state, in which the condition of the elderly is under or on the floor accidentally with the presence or without witnesses. Falls are one of the leading causes of death and injury to the elderly. About 20% to 30% of the elderly who have a high degree of disability due to falling will experience a loss of freedom in carrying out daily activities [2]. The incidence of falls in the elderly every year is recorded to be around 30% occur in the elderly aged 65 years old or older and for the elderly aged 80 years old or more fall incidence rate increases to 50% worldwide [3]. It is estimated that 646,000 people die every year as a result of falls, that makes falls the second cause of death from accidental injuries, after a traffic accident. Two factors cause falls in the elderly are, namely, intrinsic factors and extrinsic factors. The intrinsic factors of risk fall in the elderly such as gender, influence of medicine, and visual impairment. Older women are particularly prone to falling and increasing the severity of injuries. Then, the elderly who consume a lot of medicines can cause some changes in the body, such as drowsiness and loss of balance so it risks of falls. In addition, damage to visual acuity, stereopsis, contrast sensitivity, or visual field has been consistently reported to increase the likelihood of falling. The extrinsic factors are factors from outside the elderly such as the environment, walking aids, and social support. Research shows severe injuries occur in the afternoon with a percentage of 50.5%, which is caused by increasingly dimmer lighting. Early detection and improvement of risk factors can significantly reduce the rate of future falls [4].

Nowadays there have been many approaches and techniques used to detect falls. Such as wearable devices that use Inertial Measurement Unit (IMU) sensors by measuring the change in acceleration on the 3-axes and the change in the angle of pitch, row, and yaw. The drawback of the device is that it must always be used, whereas the elderly sometimes forgets to use the device and refused on the grounds of discomfort [7]. In addition, the use of cameras in fall detection is also sensitive to dark room lighting conditions and violates patient privacy [6]. Currently, there have also been several studies that use radar sensors for the detection of hand movements, heartbeats, and falls, but the final processing of data including the application of the CNN method is still carried out on computers [5][9][10]. Therefore, this research uses a radar sensor to eliminate the shortcomings in the use of the wearable and camera sensors.

In this paper, fall detection is designed by utilizing radar sensors to acquire the object movement and the Convolutional Neural Network (CNN) algorithm to classify the object movement. The doppler frequency shift signals from the radar sensor are acquired with a duration of 2.5 seconds, with a sampling frequency of 1.5 kHz. Then, the acquired signals are converted into a spectrogram with a size of 119 x 65 pixels as the input image to the CNN. The process of data acquisition, data labeling, spectrogram conversion, and CNN's model training are carried out on the Edge Impulse platform. For the implementation stage, the CNN's model is deployed in the Arduino Nano 33 BLE Sense. The doppler frequency shift signals from the radar sensor are acquired by the Arduino Nano 33 BLE Sense to be recognized and sends the prediction result to the smartphone. From the application built in the smartphone, the information of the detected object under the fall detection system can be monitored in a real-time.

This paper is organized into four sections. The first section describes the background and general description of the conducted research. The second section discusses the system design. The third section informs the experimental results. Finally, the fourth section concludes the results.

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II. System Overview

A. System Design

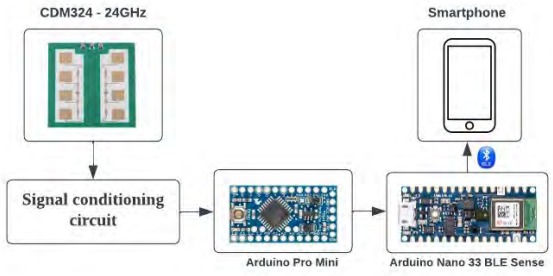


Fig. 1: System Diagram

Fig 1 illustrates the system diagram block. The radar sensor is CDM324 24-GHz. The output of the radar sensor is adjusted by the signal conditioning circuit in order to conform with the analog to digital converter (ADC) of the Arduino Pro Mini. The Arduino Pro Mini is used to handle the signal acquisition. The Arduino Pro Mini listens to the radar sensor for every 2.5 seconds as one data frame and sends the data frame to the Arduino Nano 33 BLE Sense via serial communication. The Arduino Pro Mini uses a sampling frequency of 1.5 kHz. The Arduino Nano 33 BLE Sense converts the data frame into the spectrogram and predict the recognition result. The prediction of the recognition result is sent to the smartphone via Bluetooth communication.

B. CDM324 24-GHz Sensor

The CDM324 is a 24-GHz frequency doppler module with an 80-degree azimuth and 32-degree elevation wave beam. The sensor's power consumption is 40 mA (at 5V), and the antenna design is also relatively tiny, with three header pins for simpler integration, resulting in an overall module size of 25x25x6 mm, as shown in Fig 2. The sensor is powered by a 5V battery and has a range of up to 15 meters. VCC, IF, and GND is the three pins available. Where IF-output is a doppler frequency with an amplitude ranging from -300 to 300 mV as an output.

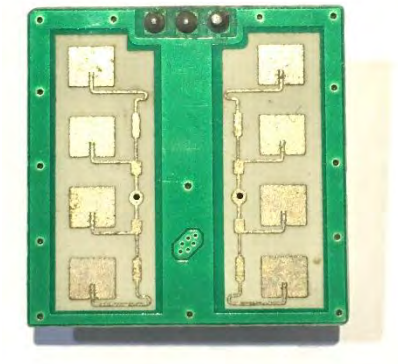


Fig. 2: CDM324 radar sensor

C. Signal Conditioning Circuit

The CDM324 radar sensor does not have an integrated set of filters and amplifiers. To be able to detect human movement from a distance, especially falling motion, it is necessary to design a signal conditioning concatenation based on sensor datasheets. The input frequency is limited to the range of 60 - 8000 Hz using a band pass filter, however, to detect human movements, it is only in the range of 40 to 500 Hz. The signal gain can be set in the range from 40 to 73 dB, which can reach the movement of objects up to 15 meters. Fig 3 shows the signal conditioning circuit. The first operational amplifier is used as the high-pass filter and the second operational amplifier is used as the low-pass filter.

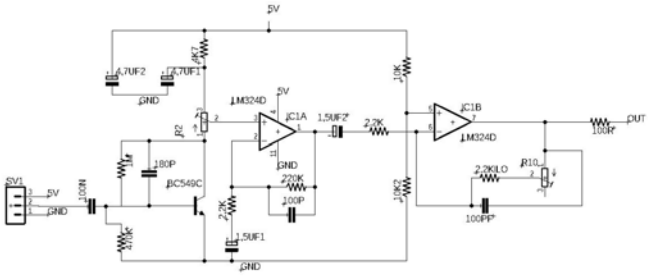


Fig. 3: Signal conditioning circuit

D. Building the Datasets

The development of the CNN in this research is shown in Fig 4, divided into two phases, namely the training phase and the testing phase. In the training phase, the collected datasets consist of several classes namely falling, vacant, and walking. Falling class means that an object is detected moving away from the radar sensor, which means the person has fallen on the floor. Vacant class means that there is no moving object under the radar sensor. Walking class means that an object is passes under the beam of the radar sensor.

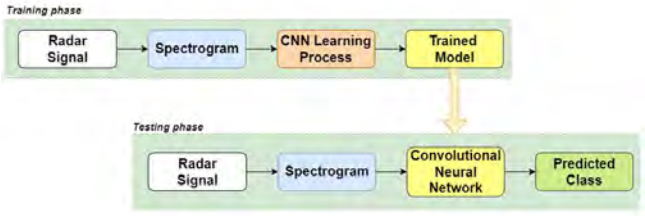


Fig. 4: Training and testing phases

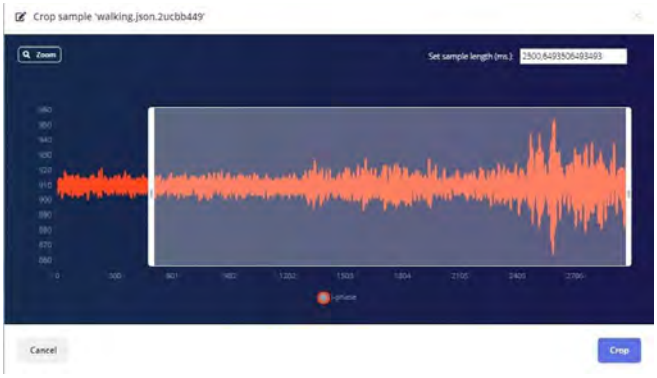


Fig. 5: Cropping the raw data as dataset

The datasets are constructed on the Edge Impulse platform as follows. Firstly, the Arduino Nano 33 BLE Sense was programmed to capture the radar signals and to send the radar signals to the PC (Personal Computer) via serial communication. In the PC side, a data forwarder program

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developed by Edge Impulse was running. The data forwarder receives the data from the selected serial port and then transmits the received data to the Edge Impulse as a raw data. Secondly, the raw data is labelled manually by hand. Fig 5 shows the labelling process by cropping the raw data for about 2.5 seconds. The collected datasets in this research is 270 samples. Finally, from 270 samples, the 216 samples are used to train the CNN's model for all classes and 54 samples are used for the validation.

E. Features Extraction

The objective of feature extraction is to transform the signal into a representative which enhances the recognition performance. For this purpose, the radar signals are converted into spectrograms. The radar signals with a duration of 2.5 seconds are segmented into overlapping frames of 200 milliseconds with a stride of 100 milliseconds. Then 119 frames will be formed. Those frames are processed using 128 FFT size. The results are spectrogram images with a size of 119 x 65 pixels. Fig 6 and Fig 7 are an example of a signal of a falling person and its spectrogram.

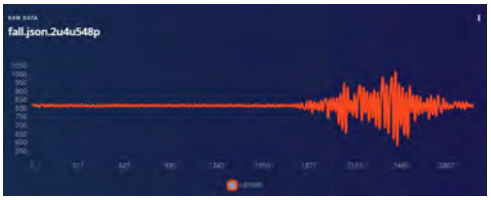


Fig. 6: Signal of falling person

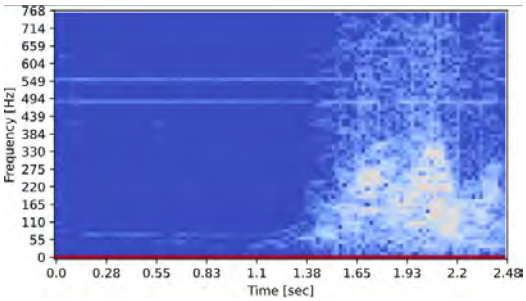


Fig. 7: Spectrogram of falling person

F. Convolutional Neural Network

The CNN classifier architecture is shown in Fig 8. The CNN consists of 2 convolution layers with max-pooling layer and dropout layer among them. The input is a 2-dimensional array of 119 x 65 pixels. The convolution process is carried out in the first layer with 8 filters and the second layer using 16 filters. The max-pooling layer serves by reducing the number of parameters from the convolution process by taking the largest value at a predetermined filter size. The dropout layer is to set random input units to zero with a frequency of 0.25 (this rate is defined) at every step during training. It means that one in every four input units is randomly dropped. At the flatten layer, the 2-dimensional array processed in the previous layer is converted into a 1-dimensional array, to be processed on a dense layer using the SoftMax function to produce a falls prediction.

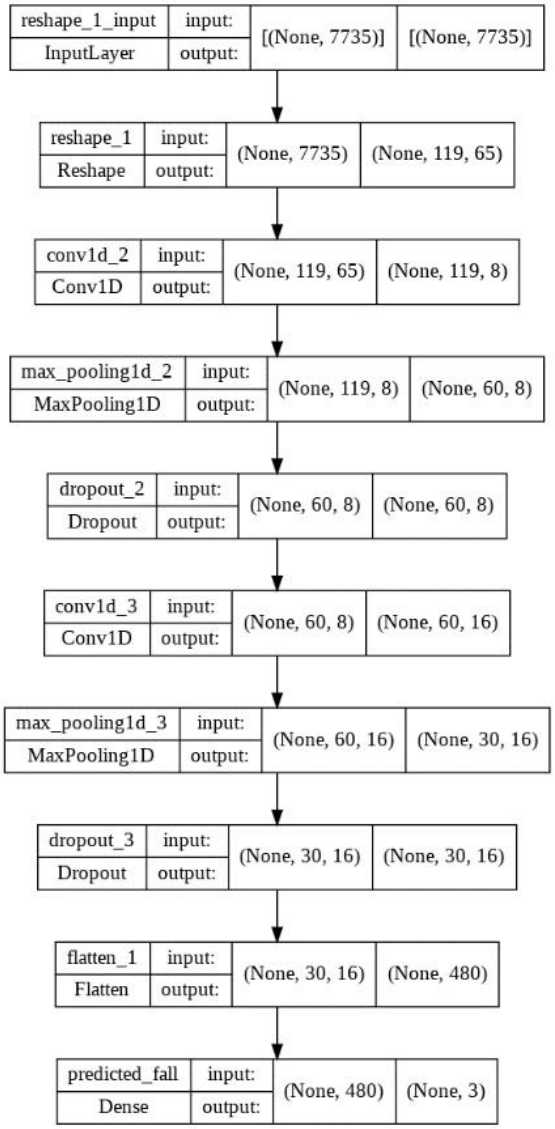


Fig. 8: CNN architecture

III. EXPERIMENTAL RESULT

The training phase has been done in the Edge Impulse. For the deployment stage, the trained model is implemented in the Arduino Nano 33 BLE Sense. The firmware generated by the Edge Impulse is utilizing Tensor Flow Lite. The Tensor Flow Lite is incapable of training models and is only capable of executing inferences. The firmware to be uploaded into the Arduino Nano 33 BLE Sense requires 374,504 bytes of program storage space and 87,384 bytes of dynamic memory. The Arduino Nano 33 BLE Sense has a flash memory of 1 Megabytes and SRAM of 256 kilobytes.

The system is implemented in a room with a dimensions of 5 x 3 x 3 meter. The system is located on the ceiling at the center of the room with the sensor position facing down, as illustrated in Fig 9. For the experimental set-up as shown in Fig 10, a thin mattress is laid out on the floor so people can demonstrate the movement of walking and falling safely under the sensor.

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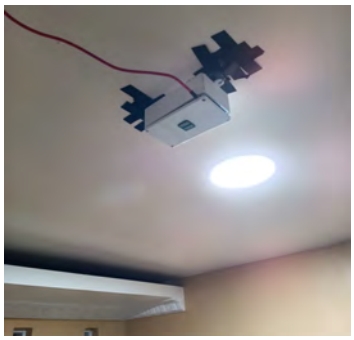


Fig. 9: Sensor placement



Fig. 10: Experimental set-up

Fig 11, Fig 12, and Fig 13 show the spectrograms of falling person, vacant, and walking person.

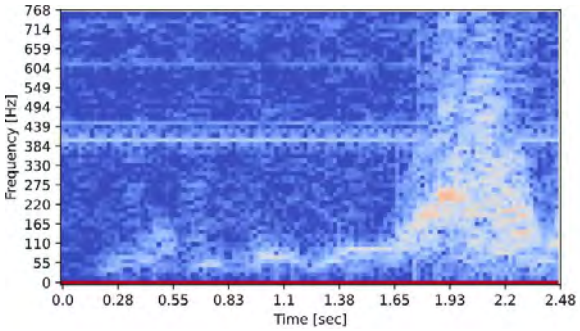


Fig. 11: Spectogram of falling person

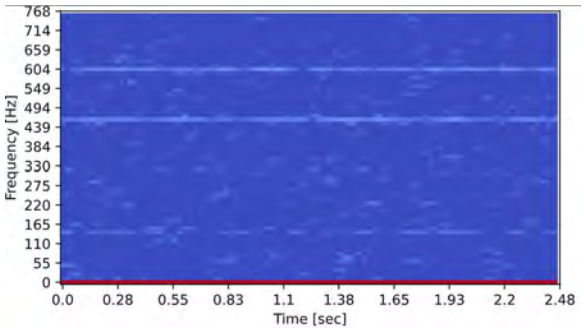


Fig. 12: Spectrogram of vacant

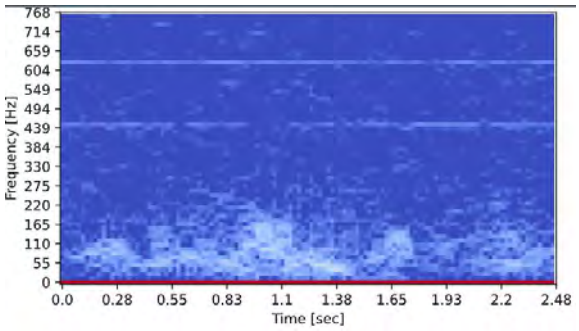


Fig. 13: Spectrogram of walking person

In the training phase, there were 270 data samples for all classes with 216 data samples for the training phase (each class has 72 samples) and 54 data for the testing phase during data validation of all classes (each class has 18 samples). The CNN's model has obtained an accuracy of 98.1% and a loss of 0.08. The confusion matrix of the training phase is shown in Table 1.

TABEL I: Confusion Matrix (Validation Set)

	Falling	Vacant	Walking
Falling	94.4%	0%	5.6%
Vacant	0%	100%	0%
Walking	0%	0%	100%
F1-score	0.97	1.00	0.97

The system has been tested using 18 new data in each class, bringing the total to 54 test data. Based on the confusion matrix shown in Table II, the overall accuracy is 96.30%.

TABEL II: Confusion Matrix (Testing Result)

	Fall	Noise	Walking	Uncertain
Fall	17	0	1	0
Noise	0	18	0	0
Walking	0	0	17	1
F1-score	0.97	1.00	0.94	-

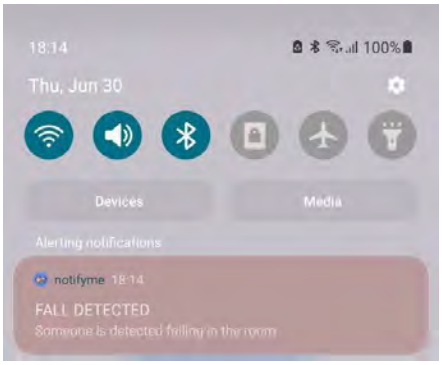


Fig. 14: The notification in the smartphone when a falling person is detected

When the CNN outputs a prediction of falls, it triggers the smartphone to pop a notification up to inform the user. Fig 14 shows the notification in the smartphone. The

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application running in the smartphone also records the CNN's prediction activities of falling person and walking person while vacant condition is not reported. The report generated by the application in the smartphone to the user is shown in Fig 15. During the system operating, it consumes a current of 56,2 mA to 59,6 mA at 5 Volts working voltage.



Fig. 15: The report generation from the CNN's prediction

IV. CONCLUSION

In this research, the fall detection using 24-GHz radar sensor for the elderly home support using CNN was proposed. The system was implemented and tested through experimentation with three classes. The radar signals are sampled with a duration of 2.5 seconds then are converted into spectrograms with a size of 119 x 65 pixels for the CNN that requires 2-dimensions as the input data. The experimental results showed an accuracy rate of 98.1% in the training phase and 96.30% in the testing phase. When the CNN generates a prediction of falling or walking, a notification to the user will be displayed.

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